Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18RA54]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER–2021

ADVANCED MICROCONTROLLERS

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1.	**a.**	Define addressing mode. Mention the various types of addressing modes in 8051 and explain with an example for each.	**08**
	b.	Explain Program Status Word (PSW) register with each flag bits in 8051.	**06**
2.	**a.**	Draw the ARM Current Program status register (CPSR) and mention the significance of each bit field according to bit position.	**06**
	b.	Outline the ARM memory organization and explain the types with neat diagrams.	**08**
3.	**a.**	Draw and address the memory mapping for LPC 2388 microcontroller, specify the address range for each location.	**07**
	b.	Illustrate the single and multiple register data transfer instructions with example in ARM state.	**07**
		OR	
4.	**a.**	Write a program to display "Hello world" using SWI instruction of ARM state.	**07**
	b.	List and explain the arithmetic instructions available in ARM state and give example for each.	**07**
5.	**a.**	Draw the Thumb instruction decompressor organization and explain the implementation for the same with an instruction example.	**09**
	b.	Mention the properties and applications of Thumb system.	**05**
		OR	
6.	**a.**	Mention the various types of power control modes in LPC2388 and highlight the features in each mode.	**07**
	b.	Explain the Thumb programmer's model and mention the available registers with their function.	**07**
7.	**a.**	List the features of Digital to Analog converter (DAC) used in LPC2388 controller	**06**
	b.	Mention the exceptions according to execution priorities in ARM and analyze the sequence of occurrence and execution of multiple exceptions.	**08**
		OR	
8.	**a.**	Mention the features of LPC2388 GPIOs and Illustrate the functions of the IODIR, IOCLR and IOSET pins in LPC2388 GPIOs.	**08**
	b.	Categorize the ARM exceptions based on their source of generation and explain the same.	**06**